

YUNFEI XIE

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RESEARCH INTERESTS

Robot learning from human video, post-training for foundation models

EDUCATION

Rice University

09/2025-2029 (expected)

Ph.D. in Computer Science, Advisor: Prof. Chen Wei

Huazhong University of Science and Technology

09/2021-06/2025

Bachelor of Engineering in Artificial Intelligence

SELECTED PUBLICATION LIST (GOOGLE SCHOLAR)

Robot Learning

1. **RoboTok: An Internet-Scale Data Engine for Human Demonstration Retrieval and Dexterous Manipulation Learning**

Howard H. Qian, Yiting Chen, **Yunfei Xie**, Kejia Ren, Podshara Chanrungrmaneeekul, Gaotian Wang, Bowen Wen, Chen Wei, Kaiyu Hang

Preprint (under review), 2026. [link]

2. **Sensorimotor Alignment: A Training-Free Predictor of Robot Policy Success**

Yunfei Xie, Howard H. Qian, Shiyi Lan, Kaiyu Hang, Chen Wei

Preprint (under review), 2026. [link]

LLM Post-training

1. **Play to Generalize: Learning to Reason Through Game Play**

Yunfei Xie, Yinsong Ma, Shiyi Lan, Alan Yuille, Junfei Xiao, Chen Wei

ICLR 2026. [link]

2. **How Off-Policy Can GRPO Be? μ -GRPO for Efficient LLM Reinforcement Learning**

Minghao Tian, **Yunfei Xie**, Chen Wei

Preprint (under review), 2026. [link]

Data Pipelines & Synthesis

1. **MedTrinity-25M: A Large-scale Multimodal Dataset with Multigranular Annotations for Medicine**

Yunfei Xie, Ce Zhou, Lang Gao, Juncheng Wu, Xianhang Li, Hong-Yu Zhou, Sheng Liu, Lei Xing, James Zou, Cihang Xie, Yuyin Zhou

ICLR 2025. (400+ GitHub stars) [link]

Computer Vision

1. **From Pixels to Objects: A Hierarchical Approach for Part and Object Segmentation Using Local and Global Aggregation**

Yunfei Xie, Cihang Xie, Alan Yuille, Jieru Mei

ECCV 2024. [link]

2. **Story-Iter: A Training-free Iterative Paradigm for Long Story Visualization**

Jiawei Mao, Xiaoke Huang, **Yunfei Xie**, Yuanqi Chang, Mude Hui, Bingjie Xu, Zeyu Zheng, Zirui Wang, Cihang Xie, Yuyin Zhou

ICLR 2026. (900+ GitHub stars) [link]

Self-Improving Agents

1. **MEMO: Memory-Augmented Model Context Optimization for Robust Multi-Turn Multi-Agent LLM Games**

Yunfei Xie, Kevin Wang, Bobby Cheng, Jianzhu Yao, Zhizhou Sha, Alexander Duffy, Yihan Xi, Hongyuan Mei, Cheston Tan, Chen Wei, Pramod Viswanath, Zhangyang Wang

ICML 2026. [link]

PROFESSIONAL EXPERIENCE

Research Intern, Adobe Research

05/2026-08/2026

Advised by Dr. Branislav Kveton

- **RL for Flow Matching:** Developed a reinforcement learning algorithm for post-training flow-matching generative models, applied to Adobe's Firefly foundation model

Research Assistant, Rice University

02/2025-

Advised by Prof. Chen Wei; robot learning in collaboration with Prof. Kaiyu Hang's RobotPI Lab

- **RoboTok:** Built an internet-scale data engine that indexes 100,000 web video clips and retrieves relevant human demonstrations by matching 3D hand motion trajectories, turning web video into training data that improves dexterous-hand RL policies on an Isaac Gym manipulation benchmark
- **Sensorimotor Alignment:** Proposed a training-free offline metric that measures how well a robot model's visual representation aligns with the geometry of its motion trajectories, predicting task success for VLAs, VLMs, and world-action models in simulation and on real robots without any policy rollouts
- **Play to Generalize:** Showed that multimodal LLMs gain transferable reasoning skills from RL post-training on simple arcade games, improving out-of-domain math and 3D spatial reasoning without ever seeing worked solutions or diagrams
- **μ -GRPO:** Showed that GRPO can train on much staler rollouts than assumed, enabling an RL recipe that alternates between a few large generation and optimization stages and matches standard GRPO at much lower training cost
- **MEMO:** Developed a self-play framework that optimizes an LLM agent's inference-time context via a persistent memory bank and tournament-style prompt evolution, raising win rates and stabilizing outcomes in multi-agent games

Research Intern, VLAA Lab, University of California, Santa Cruz

12/2023-02/2025

Advised by Prof. Yuyin Zhou and Prof. Cihang Xie

- **MedTrinity-25M:** Built a scalable data synthesis and labeling pipeline that grounds, annotates, and filters raw medical images without any paired text, producing the largest multimodal medical dataset, 25 million images across 10 modalities, and state-of-the-art models when used for pretraining
- **Story-Iter:** Designed a training-free iterative method for visualizing long stories that refines each frame against all previously generated frames, achieving state-of-the-art consistency across long sequences

Research Intern, CCVL Lab, Johns Hopkins University

07/2023-12/2023

Advised by Prof. Alan Yuille

- **From Pixels to Objects:** Developed a hierarchical vision transformer that groups pixels into superpixels for part-level detail and superpixels into larger groups for object-level context, achieving state-of-the-art part and object segmentation

HONORS AND AWARDS

Lambda Research Grant (\$10,000)

08/2025

Science and Technology Scholarship, Huazhong University of Science and Technology (top 2% in school) 2022

INVITED TALKS

"Play to Generalize: Learning to Reason Through Game Play", Open AGI Symposium, NeurIPS 2025 12/2025

ACADEMIC SERVICE

Reviewer:

2026: ICLR, ICML, CVPR, ECCV

2025: ICCV, ICLR, CVPR, ICML, TMM, NeurIPS, TPAMI

2024: CVPR, ICML, IEEE ISBI

PROFESSIONAL SKILLS

ML Frameworks: PyTorch, verl, OpenRLHF, vLLM, DeepSpeed, FSDP, Slurm

Robotics: ROS, MuJoCo, NVIDIA Isaac Gym, SAPIEN

Languages & Tools: Python, Shell, Git, Docker, Linux